



AadhaarLens

Unlocking societal trends in Aadhaar enrolment & update activity — where new registration ends and lifelong maintenance begins.

Our angle: Aadhaar today is overwhelmingly a **maintenance system, not a registration system** — for every new enrolment recorded, the network processes roughly ten update transactions. We map where that maintenance load concentrates, which districts run "hot," and what that means for how UIDAI should staff and prioritise its centres.

92,028

CLEAN ACTIVITY
RECORDS ANALYSED

38 / 875

STATES & UTs / DISTRICTS
COVERED

9.64×

UPDATE-TO-ENROLMENT
RATIO

4

DISTRICT BEHAVIOURAL
PERSONAS FOUND

Members

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Dataset window

01 Apr – 19
Sep 2025

GitHub: github.com/pawano6/UIDAI-AadhaarLens · Deliverable:
Consolidated PDF Report

01 Problem Statement & Approach

What we were asked, and the specific question we chose to answer

Aadhaar enrolment and update activity varies across age groups, states, districts, and activity types. Our objective was to identify meaningful patterns in new-enrolment and Aadhaar-maintenance activity, detect differences between regions, flag unusual district-level activity, and translate all of this into operational recommendations UIDAI could act on next quarter.

Our chosen angle: rather than another "enrolments by state" bar chart, we treat every row as a unit of *operational workload* and ask where that workload is registration-driven versus maintenance-driven — because the two require completely different staffing, kits, and outreach.

Q1 Which age groups drive new enrolment?
And how does that mix shift across states?

Q2 Do demographic and biometric updates move together?
Or do they diverge by geography, revealing different maintenance regimes?

Q3 Which districts are enrolment-heavy vs. maintenance-heavy?
Volume alone hides which districts are under real operational pressure.

Q4 Can districts be grouped into behavioural personas, and can unusual activity be flagged?
To move from state-level averages to a district-level targeting framework.

Analytical path followed: **Data Cleaning** → **Feature Engineering** → **Exploratory Analysis** → **District Segmentation** → **Anomaly Detection** → **Recommendations**.

02 Datasets Used

Source, scope, and the columns our analysis is built on

We used the **UIDAI Aadhaar Enrolment & Update Records** dataset provided for the hackathon exclusively — no external data was joined in, so every number in this report traces back to this single source. Each row is an **aggregated activity count** for one date × one geography (state → district → pincode), broken down by age band and activity type. Rows represent activity volume, not individual applicants.

Column	Type	What it represents
date	Date	Calendar date the enrolment/update activity was recorded
state / district / pincode	Text	Geography of the enrolment centre, down to 6-digit PIN code
age_0_5, age_5_17, age_18_greater	Numeric	New-enrolment counts by age band
demo_age_5_17, demo_age_17_	Numeric	Demographic updates (name, address, gender, mobile, DOB) by age band
bio_age_5_17, bio_age_17_	Numeric	Biometric updates (fingerprint / iris / photo re-capture) by age band

94,955 → 92,028

RAW ROWS AFTER REMOVING 2,927 EXPORT DUPLICATES (3.08%)

17,560

UNIQUE PINCODES, ALL VALIDATED TO 6-DIGIT FORMAT

23 DAYS

DISTINCT ACTIVITY DATES, 01 APR – 19 SEP 2025

03 Methodology

How the raw export became an analysis-ready dataset

We treated cleaning as evidence, not an assumption — every step below was checked against the raw data before we trusted a single chart.

Data cleaning

- Dropped the exported `Unnamed: 0` index column — this alone was responsible for a false "0 duplicates" reading.
- Re-checked duplicates on real columns only: found and removed **2,927 true duplicate rows (3.08%)**.
- Confirmed **zero missing and zero negative values** across all seven activity columns — but flagged that a large share of cells are legitimate zeros (e.g. 91,474 rows with zero adult enrolment), which we treat as "no activity," not "not recorded."
- Converted `date` to proper datetime; validated the range (01 Apr – 19 Sep 2025, 23 distinct dates).
- Standardised state names (e.g. *Orissa* → *Odisha*, *Pondicherry* → *Puducherry*) and title-cased district names to prevent silent under/over-counting on group-by.
- Validated pincodes to a strict 6-digit, valid-first-digit pattern — 100% of the cleaned rows passed.

Feature engineering

- `total_enrolment` — sum of all three age bands
- `total_demo_updates` / `total_bio_updates` / `total_updates`
- `update_to_enrolment_ratio` — total updates ÷ total enrolment; our core "maintenance pressure" signal
- `child_share` — age 0–5 enrolment ÷ total enrolment
- `biometric_share_of_updates` — biometric ÷ total updates, used to separate demo-heavy vs. bio-heavy regions
- `year / month / weekday / is_weekend` — calendar features for temporal checks

Analytical techniques applied

- Univariate distributions of activity composition and age mix
- Bivariate state-level comparison of age mix and update-type mix
- District-level scatter of enrolment volume vs. update ratio (log-log)
- **K-Means clustering** (k=4, standardised features) to build district behavioural personas from enrolment, update ratio and child share
- **Z-score anomaly detection** per district on September daily enrolment, flagging $|z| \geq 3$

Before / after — why the index column mattered: a naive `df.duplicated().sum()` returned **0** duplicates, because every exported row carried a unique index number. Only after dropping `Unnamed: 0` did the true figure surface: **2,927 duplicate activity records (3.08% of the file)**. Skipping this step would have quietly inflated every downstream total.

Sample cleaning code (from the working notebook)

```
df = df.drop(columns=['Unnamed: 0']) # remove export artefact
df = df.drop_duplicates() # 94,955 → 92,028 rows
df['state_clean'] = df['state'].replace(state_mapping) # e.g. Orissa → Odisha
df['district_clean'] = df['district'].str.strip().str.title()
df['pincode'] = df['pincode'].str.replace(r'\.0$', '', regex=True).str.zfill(6)
df['update_to_enrolment_ratio'] = np.where(
    df['total_enrolment'] > 0, df['total_updates'] / df['total_enrolment'], np.nan)
```

04 Data Analysis & Visualisation

What the numbers show, one chart at a time

598,001

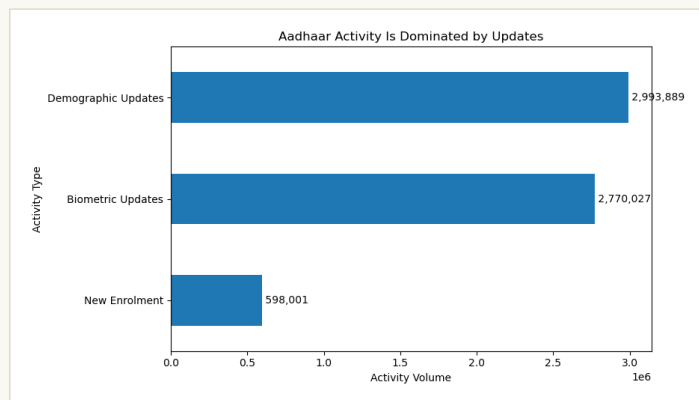
TOTAL NEW-ENROLMENT ACTIVITY

5,763,916

TOTAL UPDATE ACTIVITY (DEMO + BIOMETRIC)

9.64×

UPDATES PER UNIT OF NEW ENROLMENT



FINDING 01 · ACTIVITY COMPOSITION

Aadhaar activity is update-heavy, not enrolment-heavy

Across the full window, demographic updates (2,993,889) and biometric updates (2,770,027) together dwarf new enrolment (598,001) — a combined update-to-enrolment ratio of **9.64×**.

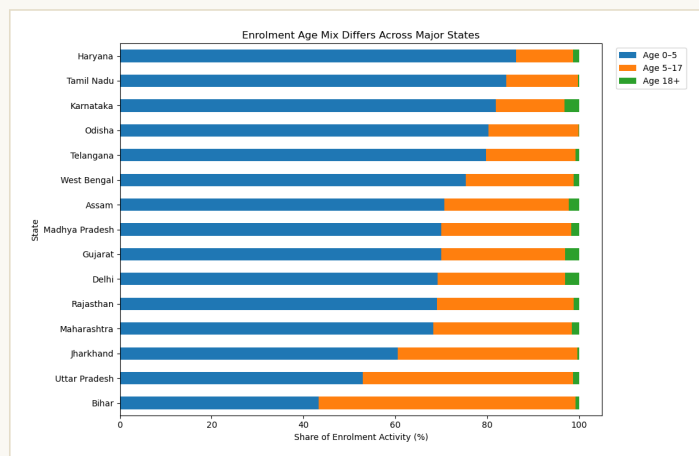
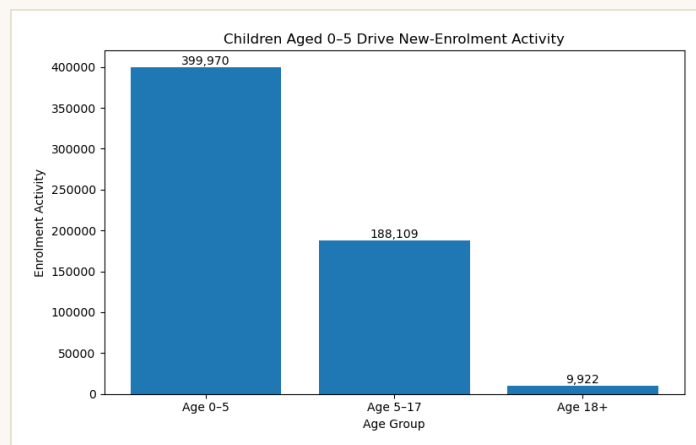
So what: the network's day-to-day workload is dominated by maintaining existing IDs, not creating new ones — centre design and staffing should be planned around update throughput first.

FINDING 02 · AGE COMPOSITION

New enrolment is overwhelmingly a child activity

Age 0–5 accounts for **66.9%** of all new enrolment (399,970 of 598,001), age 5–17 for 31.5%, and adults (18+) for just **1.7%** — almost all adults in this window are already enrolled and are updating, not enrolling.

So what: front-line enrolment infrastructure (child-friendly kits, anganwadi/school tie-ups) matters more for growth than adult-facing enrolment drives.

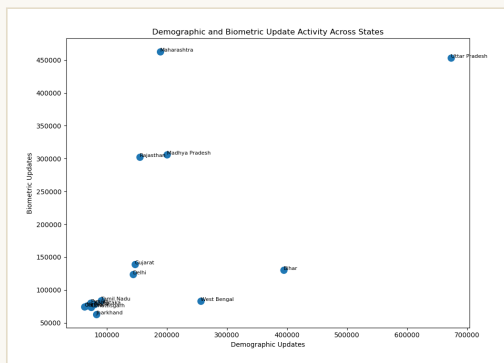


FINDING 03 · GEOGRAPHIC VARIATION

The enrolment age-mix is not uniform across major states

Haryana (86.3%), Tamil Nadu (84.1%) and Karnataka (81.9%) skew heavily toward age 0–5, while Bihar is the outlier — just 43.3% age 0–5 versus **55.9% age 5–17**, the highest older-child share of any large state.

So what: a single national enrolment-drive template will under- or over-serve specific states — Bihar in particular needs an older-child (5–17) enrolment push, not just a newborn one.



FINDING 04 · UPDATE-TYPE GEOGRAPHY

States diverge on demographic vs. biometric update mix

Maharashtra is the most biometric-skewed major state (71.0% biometric), while Bihar and West Bengal run the opposite profile (~75% demographic).

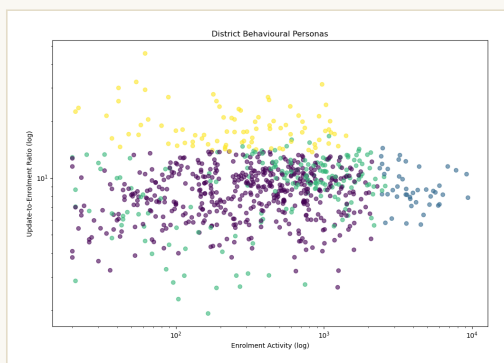
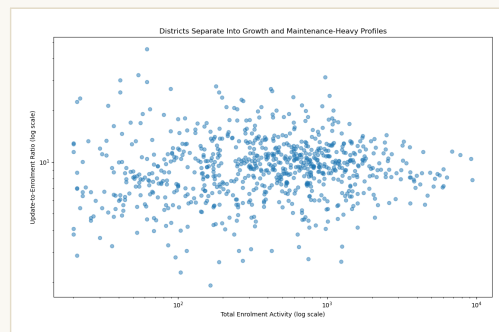
So what: biometric-capture capacity should go to Maharashtra-like states, demographic-correction capacity to Bihar/West Bengal-like states.

FINDING 05 · DISTRICT OPERATING PROFILES

High enrolment volume does not mean high update pressure

Enrolment volume vs. update ratio (log-log) shows two genuinely different populations of districts — high-volume ones aren't automatically the highest-maintenance ones. The most extreme ratios (Manipur districts, 30-45x) come from very low-volume centres.

So what: the ratio, not the raw total, should drive resourcing decisions.



FINDING 06 · DISTRICT SEGMENTATION

Districts fall into four behavioural personas

K-Means (enrolment, update ratio, child share) separates districts into four clusters: **high-volume/low-ratio "growth engines,"** **low-volume/high-ratio "maintenance-heavy,"** and two mid-volume groups split by how child-driven enrolment is.

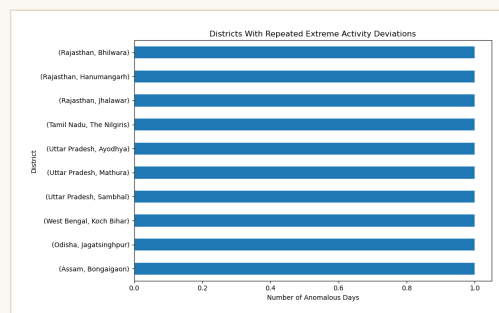
So what: a ready-made targeting framework beyond state-level averages.

FINDING 07 · ANOMALY DETECTION

September anomalies are one-off spikes, not repeated problems

20 district-days crossed $|z| \geq 3$ in September, topped by Mysuru, Karnataka ($z=3.73$) and Koch Bihar, West Bengal ($z=3.63$). Every one of the top 10 flagged districts shows exactly **one** anomalous day, not a repeated pattern.

So what: these read as one-time events (a camp, a drive) worth a quick manual check, not a structural fix.



05 Recommendations & Impact

If we were UIDAI, here is what we would do differently next quarter

- 1 **Differentiate resource allocation by district, not by state average.** Enrolment volume and update pressure are largely independent — a district-level, not state-level, staffing model is needed.
- 2 **Prioritise child-focused enrolment capacity** in high child-share districts (age 0–5 drives 66.9% of national enrolment) — kits, outreach and anganwadi/school linkages should scale with this, not with adult-enrolment assumptions.
- 3 **Match update capacity to update type, regionally.** Give biometric-capture capacity to Maharashtra-like, biometric-skewed states, and demographic-correction capacity to Bihar/West Bengal-like, demographic-skewed states.
- 4 **Route the four district personas into a planning tool.** "Growth engine" districts need enrolment throughput; "maintenance-heavy" districts need update/renewal capacity — a two-track operating model beats one national playbook.
- 5 **Treat flagged anomalies as a manual-check queue, not a policy trigger.** Every top-10 September anomaly was a single-day event — route these to a lightweight verification step (camp record, data-entry check) before acting on them.

Impact & applicability: every recommendation above is directly actionable from data UIDAI already collects — no new data source is required, only a shift from state-level dashboards to a district-persona-and-ratio view.

06 Limitations

What we are, and are not, claiming

The dataset holds **aggregated activity counts**, not person-level records — we cannot track an individual across enrolment and later update events.

The **update-to-enrolment ratio** is an activity-volume indicator, not a measure of how often any one individual updates their record.

The available dates (23 days across a 6-month window) are **not a continuous long-term series**, so we avoid strong seasonal or trend claims.

Detected anomalies are **candidates for investigation**, not confirmed causes — we propose plausible explanations (camps, drives, data entry) without asserting them.

Code & full notebook

The complete working notebook — cleaning, feature engineering, every chart above, the K-Means segmentation, and the z-score anomaly scan — is available in full at: github.com/pawano6/UIDAI-AadhaarLens